

COMPARISON OF HYBRID FEM/TSEA AND EXPLICIT FEM NUMERICAL METHODS FOR MECHANICAL SHOCK MODELLING AND TEST SYSTEM DEPENDENCE IN SHOCK SOURCE CHARACTERIZATION

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ABSTRACT

Predicting mechanical shock levels before testing and flying is an important objective for space systems. Current modelling methods are not well-established in the space sector, necessitating the study and comparison of different approaches. The first contribution is the comparison of two methods: explicit FEM and hybrid FEM/TSEA. The former is well-known but not sufficiently effective for studying shock tests on complex space systems, while the latter is getting significant improvements in recent years, but without defined usage for shocks in the space sector. Ultimately, explicit FEM appears efficient and precise for simple systems, whereas hybrid FEM/TSEA shows better potential in complex systems like satellites.

The second contribution of the article shows the shock source characterization system's dependence on injection data and results from the hybrid FEM/TSEA method. Thus, normalization of the energy distribution relative to the characterization system's modal basis is required before injecting it into the studied system.

1. INTRODUCTION TO MECHANICAL SHOCKS

1.1. Context of mechanical shocks in satellite structures

Space systems must meet mission requirements despite operational constraints. Mechanical shocks, from launcher events (e.g., stage separations, fairing release) and satellite events (e.g., separation, deployment), are critical specifications. These shocks are mainly pyrotechnic or purely mechanical. To prevent failures in sensitive components like electronics and instruments, estimating resistance to expected shocks is essential. CNES Toulouse uses its own testing tools: the ESA/CNES shock bench (Fig. 1 left) for small equipment tests up to 10,000g at 1000Hz, and the portable PyroShockGun (PSG) (Fig. 1 right), which can be mounted on any system, including large spacecraft, with multiple PSGs used simultaneously. This system works by igniting a powder cartridge, which propels an impactor to strike the breech, generating a shock. The resulting shock varies significantly depending on the tested structure, making tests difficult to predict.



Figure 1: shock bench test ESA/CNES and MicroCarb Structural Model shock test with PyroShockGuns

Numerical methods could become the future means for predicting shock levels in spacecraft structures, but robust and accurate modelling and simulation methods are crucial. Current predictions rely on experience, empirical design rules [1, p.122], analogies from past projects [2, sec.8] and tests, as outlined in ECSS standards. Inaccurate predictions may lead to improper system qualification, so an empirical approach is often used to characterize shock sources and qualify subsystems and components. This requires system similarities or complex, costly, risky, late and often poorly controlled tests.

Consequently, improving modeling methods is essential. CNES is currently collaborating with a PhD student, the author of this work, on modeling mechanical shocks in space structures. The goal is to provide engineering recommendations for using numerical tools, enhancing accuracy, robustness, and simplicity. Key areas include shock source modelling and shock propagation through structures and interfaces.

1.2. Introduction to shock tests and numerical methods used in this study

To address these issues, various test campaigns and numerical models were conducted to correlate experimental and numerical results. Initial studies involved dropping a steel ball onto an aluminum plate, with foams simulating free boundary conditions. A parametric study showed their effectiveness. As experiments grew more complex, they included equipment dummies to study interfaces, and furthermore shock source modelling with the PSG, and impact tests on 3D plates and sandwich panels. Tests on complex

systems like MicroCarb and Yoda satellites structural models will allow us to verify the effectiveness of the engineering methods developed.

Various numerical methods are used in impact studies, chosen based on modelling complexity, and desired robustness and accuracy. Three main criteria for selecting numerical tools are model complexity, type of shock injection, and frequency domain. The frequency ranges can be defined as follows: **Low frequencies:** resonant behavior with natural frequencies and distinct peaks.

High frequencies: transient behavior dominates, with overlapping modes smoothing the response.

Mid-frequencies: WEISSER [3] defines it simply as the intermediate region between high and low frequencies. GAGLIARDINI et al. [4] define it as the region where the structure exhibits neither modal nor transient behavior. The frequency limit values vary based on the type of structure being analyzed. However, for spatial systems such as satellites, the ECSS defines the mid-frequency range between 200 Hz and 2000 Hz [2, p.157]. According to NASA standards [1,p.107], the accuracy of finite element models (FEM) decreases beyond the 50th mode of a structure, which can be seen as the threshold between low and high frequencies. This mode could be considered as the central value for mid-frequencies.

Implicit FEM is efficient for low-frequency resolution, while explicit FEM handles higher frequencies [1,p.106][5]. Initial parametric studies with explicit FEM provide preliminary recommendations for modelling mechanical shocks in aerospace structures [6].

2. INTRODUCTION TO HYBRID FEM-TSEA

Statistical Energy Analysis (SEA), developed in the late 1950s and refined by Richard H. Lyon [7], is used for high-frequency system analysis with energy exchanges calculation methods between subsystems. SEA or its variants can be preferred for preliminary design [1, p. 122]. Nevertheless, SEA can be a complement for high-frequency predictions, when FEM is used to predict low frequency levels [1, p. 183].

However, SEA relies on several assumptions [8]:

- Each subsystem has a finite number of modes.
- Modes are equally probable within the considered frequency range.
- Total energy within each subsystem is evenly distributed among its modes.
- Modal amplitudes are assumed to be incoherent, meaning that the contributions of each vibration mode to the total structural response are statistically independent, resulting in random rather than systematic constructive or destructive interactions.
- Damping is uniform across modes (although not strictly necessary, but it simplifies the method).
- The diffusion is ideal, and the coupling is weak [2]

Classical SEA has limitations as it primarily predicts responses in steady-state conditions [9]. Methods like Transient SEA (TSEA) [10] have been developed to address this issue, introducing transient resolution by modelling the system at discrete time intervals to incorporate temporal dynamics into the power balance. TSEA assumes rapid energy transfer between excited and coupled systems, but in reality, energy propagates

through structures, requiring numerical integration over time steps relative to system traversal times [10].

Furthermore, to address the fact that the TSEA resolution is less suitable for low frequencies than FEM, as it stems from the SEA method, various hybrid FEM-SEA approaches have been proposed. Shorter, Langley, and Cotini have introduced their methods, implemented in the Wave6 software used in this study, in publications [11] [12]. Hybrid FEM-SEA allows a structure to be divided into global subsystems, handled deterministically (FEM), and local subsystems, treated statistically (SEA). Subsystems can be categorized based on the Characteristic Wavelength Method, which determines locality by the number of wavelengths within the subsystem's characteristic length [13].

Applicability of numerical methods: predictive methods versus experimental correlation methods

It's essential to differentiate purely predictive methods and those using experimental results. Predictive methods model the physical shock event or use shock source characterization, while correlative methods use test data to describe shock injection.

Predictive methods for tests involve models that directly represent shock source or use data from shock source characterization, such as spectral energy distribution. The predictive approach presented in this article for hybrid FEM/TSEA involves conducting a test campaign on the shock source (Fig. 2) in various configurations using a reference structure, which is then modelled numerically. The measured SRS results from the characterization test are used as the shock source in this model to deduce the injected energy as a function of frequency corresponding to the shock source. This data, then normalized relative to the modal basis of the reference structure, serves as input to predict the shock response of any other structure excited by the same shock source.

The explicit FEM predictive approach fully represents the shock source – the PSG system and its internal dynamics. It uses the internal temporal pressure inside the combustion chamber (independent of the tested structure) as an input to the simulation. The PSG model was correlated (damper material model) with a characterization test (Fig. 2).

These methods allow for predicting shock levels during tests.

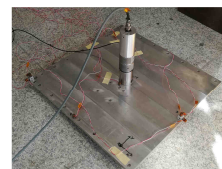


Figure 2: PSG characterization on aluminum plate

Correlation methods rely on experimental data from the same test as the one being simulated, making them inherently non-predictive. Examples include applying a measured force, using an experimental SRS or applying measured temporal accelerations for shock source modelling.

Flight shock level prediction methods can only use flight specifications provided by the launch vehicle. Since most specifications are SRS, the FEM/TSEA hybridization method implemented in Wave6 adapts well by using direct SRS injection.

3. COMPARISON OF RESULTS BETWEEN HYBRID FEM/TSEA AND EXPLICIT FEM

3.1. Simple case: A plate and a dummy

In the context of the research conducted, LS-DYNA is used for explicit FEM numerical models. Wave6, on the other hand, is used for hybrid FEM/TSEA models. The first comparative case study explored involves a ball drop on an aluminum plate with a structural dummy of space equipment. This dummy has 4 corners on which different interfaces have been tested (excess thicknesses, steel washers, permaglass washers), including direct plane/plane contact between the dummy and the plate. In this article, the comparison is limited to the case of the 4 excess thicknesses in the corners of the dummy. The interface between the plate and the dummy then consists of only 4 surfaces of 6 x 6 mm. Experimentations and modelling details are provided in the appendices.

Table 1: Percentage of numerical values in the intervals +/-3, +/-4, +/-6dB compared to experimental results and correlation coefficient for the plate and dummy case, in **Hybrid FEM-SEA / TSEA**

	+/-3dB	+/-4dB	+/-6dB	Correlation ¹
ACC1Z	73.7%	82.8%	91.5%	0.981
ACC2Z	64.6%	77.0%	86.5%	0.987
ACC3Z	53.8%	56.4%	70.8%	0.948
ACC4Z	69.9%	78.7%	92.3%	0.974
ACC5Z	53.0%	62.7%	77.4%	0.964
ACC6Z	66.0%	74.1%	89.3%	0.952
ACC7Z	76.2%	87.6%	93.3%	0.973
Mean	65.3%	74.2%	85.6%	0.9684

Table 2: Percentage of numerical values in the intervals +/-3, +/-4, +/-6dB with respect to experimental results and correlation coefficient for the plate and dummy case, in **explicit FEM**

	+/-3dB	+/-4dB	+/-6dB	Correlation ¹
ACC1Z	77.0%	90.6%	100%	0.963
ACC2Z	85.6%	93.3%	96.7%	0.964
ACC3Z	59.4%	97.8%	100%	0.971
ACC4Z	92.2%	95.9%	100%	0.981
ACC5Z	82.1%	86.8%	95.4%	0.971
ACC6Z	91.8%	96.7%	98.1%	0.937
ACC7Z	88.1%	94.8%	98.7%	0.921
Mean	86.6%	93.7%	98.4%	0.9584

From the statistical Tabs. 1, 2 and Figs. 3 and 4, it is observed that the numerical results obtained using the hybrid FEM/TSEA approach are slightly less correlated with the experimental results than the numerical results from explicit FEM. Nonetheless, even though both are simplified, the results meet the expectations. However, FEM/TSEA method remains globally less correlated at low frequencies. The discrepancy for the FEM/TSEA method may be due to the plate case not meeting the assumption that total energy is distributed across all system modes. Complex structures with irregular shapes or mixed materials often lead to better energy distribution due to internal reflections, refractions, and wave interference. Therefore, the next comparison case will address a complete satellite structure.

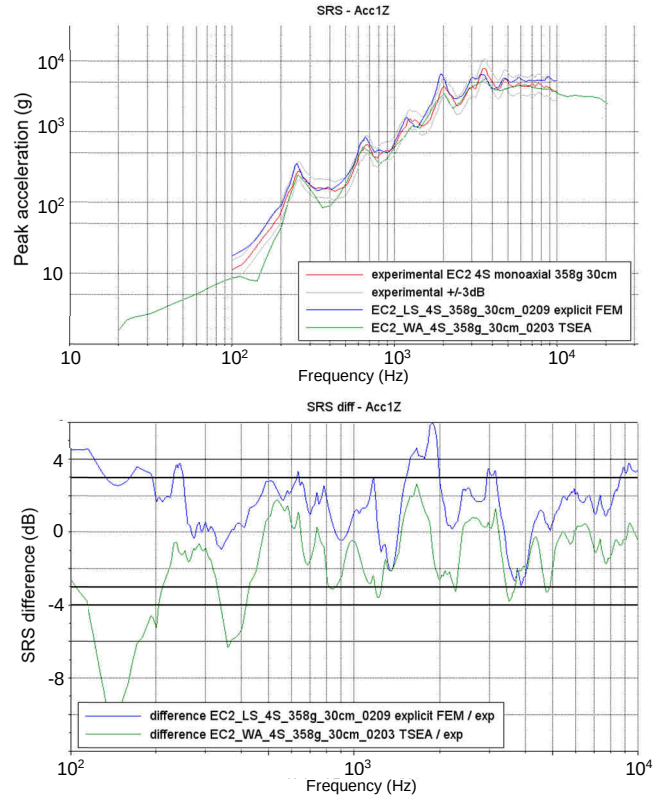


Figure 3: Comparison of ACC1Z SRS between numerical results in Hybrid FEM/TSEA, Explicit FEM, and experimental results

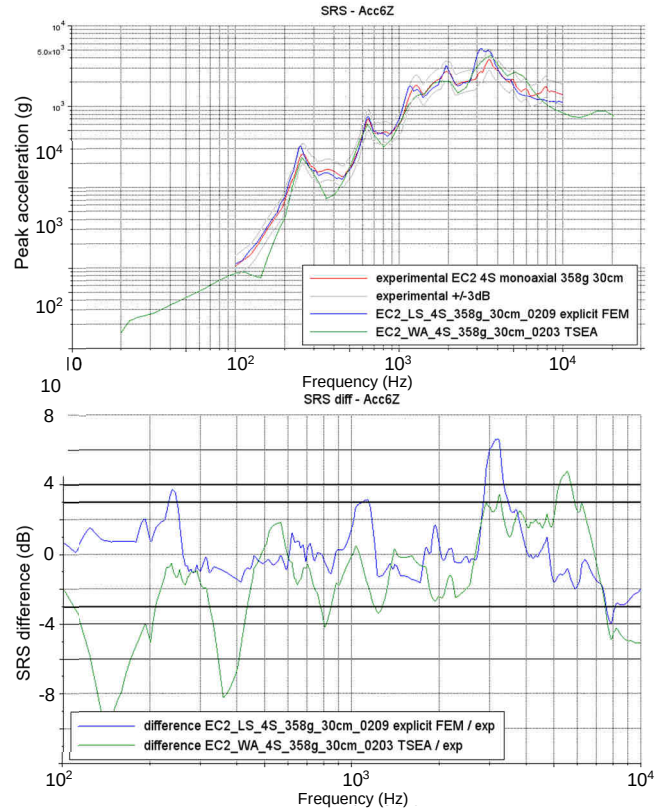


Figure 4: Comparison of ACC7Z SRS between numerical results in Hybrid FEM/TSEA, Explicit FEM, and experimental results.

¹ The correlation coefficient is the covariance between experimental and numerical results divided by the product of the standard deviations

3.2. Satellite case: Structural Model (SM) of MicroCarb

In the second case, the focus is on modelling the shock test on the structural model of MicroCarb. A photo of the test setup is provided, and important modelling parameters are listed in the appendices. The objective is to compare three methods: the correlation method using TSEA and experimental SRS injection; the predictive method using TSEA and the normalized energy distribution from shock source characterization; the predictive method using explicit FEM. For more details about the normalized energy distribution, please refer to the following section.

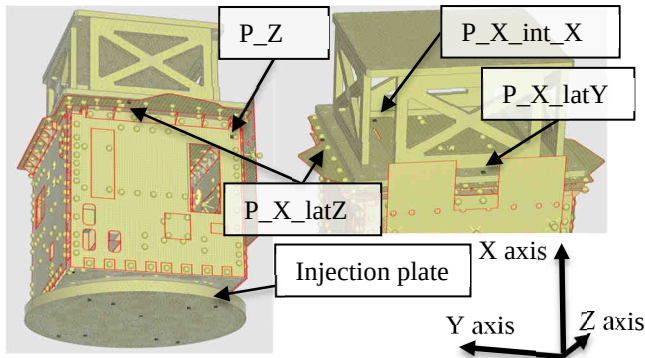


Figure 5: MicroCarb SM Illustration (Wave6)

Table 3: Percentage of numerical values within ± 3 , ± 4 , ± 6 dB of experimental results and correlation coefficient for MicroCarb Structural Model in **Hybrid FEM/TSEA**, using mean experimental SRS out-of-plane of the injection plate - T35 ECSM.

	+/-3dB	+/-4dB	+/-6dB	Correlation ¹
Mean on X injection plate	17,3%	25,3%	44,6%	0,927
Mean on Y injection plate	49,8%	62,1%	79,5%	0,887
Mean on Z injection plate	29,2%	40,0%	60,9%	0,860
P_X_int_X	14,3%	25,9%	47,2%	0,891
P_X_int_Y	46,2%	58,3%	74,9%	0,734
P_X_int_Z	39,3%	50,3%	66,6%	0,587
P_X_latY_X	47,2%	66,8%	83,1%	0,797
P_X_latY_Y	45,3%	55,6%	71,2%	0,600
P_X_latY_Z	39,2%	50,9%	69,4%	0,494
P_X_latZ_X	37,0%	42,6%	59,9%	0,723
P_X_latZ_Y	37,0%	48,6%	70,1%	0,618
P_X_latZ_Z	25,4%	43,6%	55,3%	0,585
P_Z_X	50,6%	60,3%	74,1%	0,874
P_Z_Y	47,2%	56,7%	75,9%	0,777
MAG_X	48,4%	56,4%	68,5%	0,820
SAS_X	38,9%	42,9%	65,5%	0,654
Mean	38,3%	49,2%	66,7%	0,739

At first, the results on the injection plate along the injection axis are less well correlated, likely due to the multi-directional injection method, a unique method in wave6 for injecting an SRS. Statistical analysis (Tabs. 3, 4 and 5) show that the predictive method and correlative method with hybrid FEM/TSEA method correlate better with experimental results than the explicit FEM method. Nevertheless, the levels are less correlated at the injection plate, which can be problematic as these are the levels typically adjusted to meet system

specifications during design. An example is shown on Fig. 6. However, the results for the predictive method

Table 4: Percentage of numerical values within ± 3 , ± 4 , ± 6 dB of experimental results and correlation coefficient for MicroCarb Structural Model in **Hybrid FEM/TSEA**, using **energy distribution** from PSG characterization on plate, identical to T35 ECSM, with **normalization**

	+/-3dB	+/-4dB	+/-6dB	Correlation ¹
Mean on X injection plate	12.8%	19.9%	41.8%	0.904
Mean on Y injection plate	43.6%	56.0%	74.0%	0.935
Mean on Z injection plate	19.7%	27.5%	48.0%	0.951
P_X_int_X	11.1%	16.9%	36.4%	0.726
P_X_int_Y	29.6%	43.9%	62.4%	0.589
P_X_int_Z	38.6%	46.1%	62.2%	0.426
P_X_latY_X	47.7%	63.8%	87.0%	0.695
P_X_latY_Y	28.8%	43.6%	66.9%	0.659
P_X_latY_Z	32.3%	45.1%	61.9%	0.460
P_X_latZ_X	25.1%	35.9%	61.3%	0.538
P_X_latZ_Y	28.4%	39.2%	54.5%	0.312
P_X_latZ_Z	37.3%	43.4%	55.2%	0.446
P_Z_X	40.8%	48.6%	69.8%	0.703
P_Z_Y	38.4%	55.6%	79.5%	0.888
MAG_X	30.4%	42.3%	61.3%	0.691
SAS_X	58.9%	73.7%	84.0%	0.891
Mean	32.7%	43.8%	62.9%	0.676

Table 5: Percentage of numerical values within ± 3 , ± 4 , ± 6 dB of experimental results and correlation coefficient for MicroCarb Structural Model in **Explicit FEM**, using **pressure curve** from PSG characterization on plate.

	+/-3dB	+/-4dB	+/-6dB	Correlation ¹
Mean on X injection plate	39.2%	52.1%	75.6%	0.658
Mean on Y injection plate	37.2%	48.8%	68.2%	0.883
Mean on Z injection plate	32.4%	47.2%	74.8%	0.855
P_X_int_X	25.4%	31.5%	44.2%	0.733
P_X_int_Y	28.1%	34.5%	43.9%	0.333
P_X_int_Z	36.2%	41.1%	62.6%	0.447
P_X_latY_X	16.1%	21.3%	34.8%	0.574
P_X_latY_Y	6.43%	8.62%	11.4%	0.550
P_X_latY_Z	14.9%	21.9%	32.1%	0.487
P_X_latZ_X	11.1%	20.1%	30.1%	0.814
P_X_latZ_Y	9.25%	16.0%	35.4%	0.241
P_X_latZ_Z	19.6%	26.2%	50.2%	0.279
P_Z_X	26.3%	35.3%	51.1%	0.479
P_Z_Y	1.1%	6.43%	11.0%	0.503
MAG_X	33.6%	41.4%	57.8%	0.691
SAS_X	20.8%	29.0%	41.4%	0.562
Mean	22.4%	30.0%	45.3%	0.568

using the injection of energy distribution are only slightly less well correlated than the SRS injection correlation method. This indicates a correct injection method for the predictive approach by using the normalized energy distribution thanks to shock source characterization. Examples of calculated levels for out-of-plane and in-plane accelerometers have been plotted in Figs. 7 and 8, showing better correlation with experimental results with both predictive and correlative methods with FEM/TSEA than explicit FEM method.

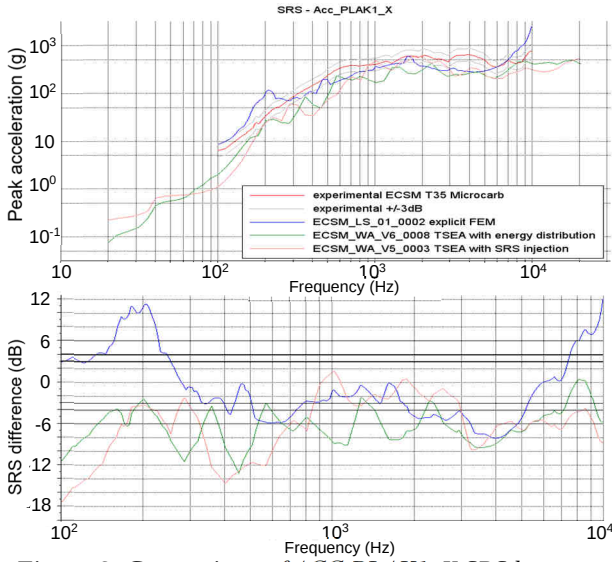


Figure 6: Comparison of ACC PLAK1_X SRS between the three numerical results and experimental results.

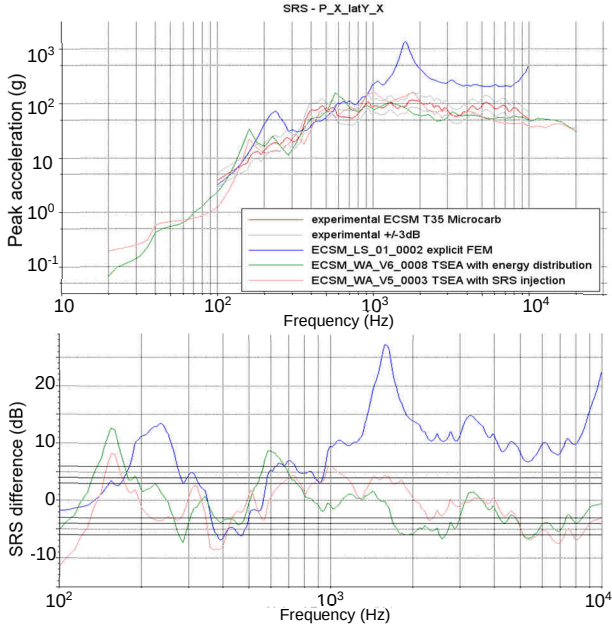


Figure 7: Comparison of ACC P_X_LATY_X SRS between the three numerical results and experimental results

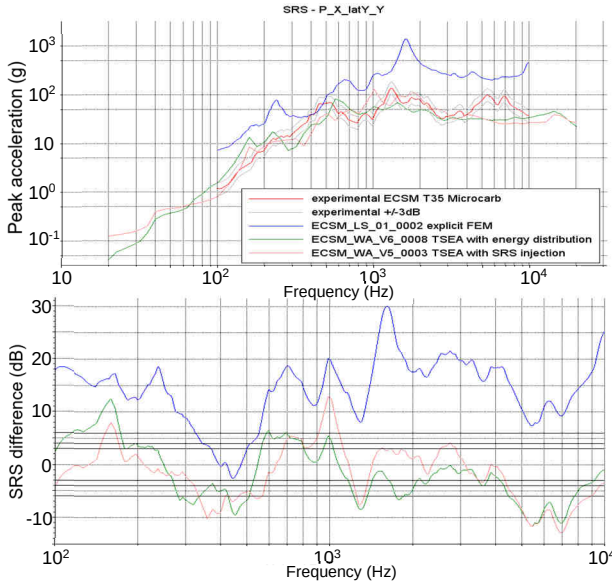


Figure 8: Comparison of ACC P_X_LATY_Y SRS between the three numerical results and experimental results.

4. THE INFLUENCE OF CHARACTERIZATION SYSTEM ON NUMERICAL RESULTS

This section demonstrates the impact of shock source characterization systems without normalizing energy distribution for the FEM/TSEA method. Key modeling parameters are in the Appendices. The example used is PSG characterization on the aluminum plate. It involves using the SRS results from PSG characterization, applying the average to the plate model, calculating the energy distribution, and normalizing it by modal density.

$$\text{normalized energy distribution}(\omega) = \frac{\text{Energy distribution}(\omega) * \text{Mass}}{\text{modal density}(\omega)}$$

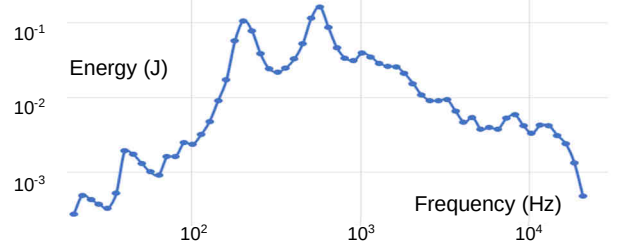


Figure 9: Frequency energy distribution from PSG characterization on plate E(ω) without normalization.

The unnormalized frequency energy distribution reveals two significant peaks corresponding to the plate's main flexural modes: around 200 Hz and 550 Hz. When this data is input into the MicroCarb Structural Model, these peaks appear in the numerical results, as shown in Figs. 10 and 11. These resonance peaks are obvious across all accelerometers without normalization. Tabs. 4 and 6 show that normalization improves correlation with experimental data by slightly reducing levels these at natural frequencies and globally lowering SRS acceleration levels. Regarding the calculated levels at the injection interface, since the levels are higher without normalization, better correlated results with the experimental data are obtained for out-of-plane accelerometers (injection direction).

Table 6: Percentage of numerical values within ± 3 , ± 4 , ± 6 dB of experimental results and correlation coefficient for MicroCarb Structural Model in **Hybrid FEM/TSEA**, using **frequency energy distribution** from PSG characterization on plate, **without normalization**

	+/-3dB	+/-4dB	+/-6dB	Correlation
Mean on X injection plate	55.7%	67.4%	81.8%	0.930
Mean on Y injection plate	29.2%	36.8%	54.2%	0.940
Mean on Z injection plate	41.3%	52.6%	67.4%	0.935
P_X_int_X	66.9%	76.3%	82.4%	0.808
P_X_int_Y	18.3%	27.4%	43.4%	0.611
P_X_int_Z	21.8%	25.5%	35.9%	0.429
P_X_latY_X	24.8%	32.3%	48.7%	0.716
P_X_latY_Y	24.8%	33.9%	54.4%	0.569
P_X_latY_Z	21.5%	29.5%	36.4%	0.319
P_X_latZ_X	32.4%	46.6%	56.0%	0.596
P_X_latZ_Y	16.9%	20.8%	35.6%	0.319
P_X_latZ_Z	22.9%	31.0%	53.8%	0.451
P_Z_X	30.6%	39.5%	58.2%	0.759
P_Z_Y	33.1%	40.1%	68.0%	0.881
MAG_X	29.9%	43.4%	69.4%	0.778
SAS_X	17.4%	23.8%	45.8%	0.837
Mean	30.5%	39.2%	55.7%	0.680

Examples are given for the injection plate on Fig. 10 and for points of interest out-of-plan and in-plan on Fig. 11.

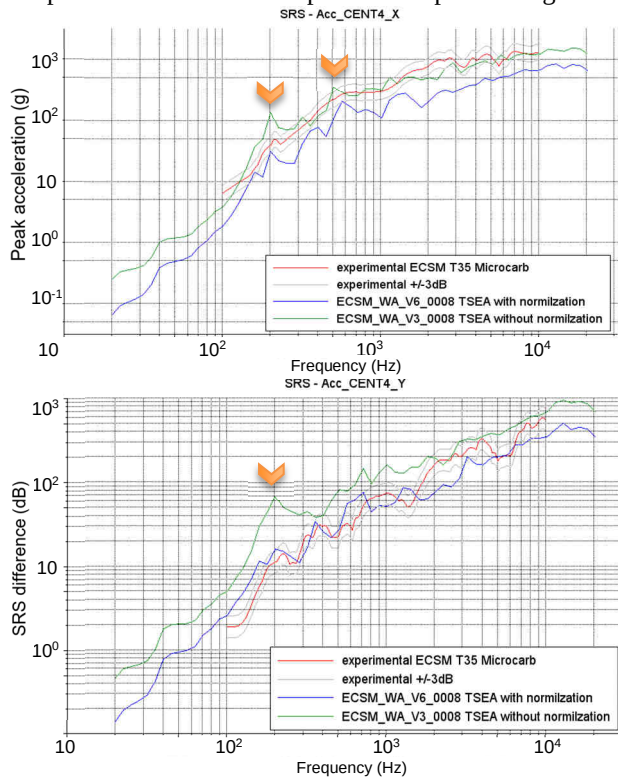


Figure 10: Comparison of ACC CENT4 SRS out-of-plan X and in-plan Y between numerical results of the models injecting the normalized and unnormalized energy distribution from PSG characterization on plate.

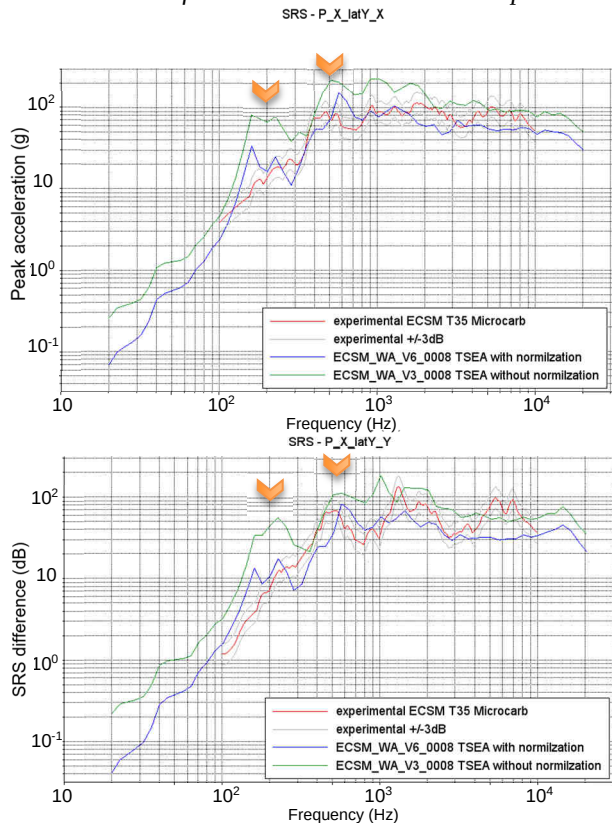


Figure 11: Comparison of ACCP_X_LATY_Y SRS out-of-plan X and in-plan Y between numerical results of the models injecting the normalized and unnormalized frequency energy distribution (PSG characterization).

5. FUTURE DIRECTIONS AND EVOLUTION OF THE PRESENTED WORK

In this study, the FEM/TSEA computational code showed promising methods and results for improving shock levels predictions for spacecrafts. Especially, normalized frequency energy distribution seems to be a convenient and accurate method for representing shock sources. These conclusions now need to be confirmed with more case studies on more complex spacecraft models – flight models instead of structural models, and the accuracy and convenience of the method for structure sizing - stresses and loads computations, should be evaluated.

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ANNEXES

Global parameters used for modelling

Table 7: acronyms and definitions of the materials used

Units	Density Kg/m ³	Young's Modulus GPa	Poisson Coefficient	Yield Strength MPa	Tangent Modulus GPa
Alu1	2.78	74.5	0.3	25	25
Alu2	2.386	74.5	0.3	300	25
Alu3	2.77	70	0.33	None	None
Steel1	7.8	195	0.3	1000	20
Steel2	7.36	193	0.3	205	19
Steel3	7.1	193	0.3	100	10
Steel4	7.8	193	0.3	None	None
Steel5	7.8	200	0.3	400	20
Ptfe	2.08	10	0.46	40	50

- For Solid Modelling: ELFORM = 2: Full integration with solid elements (8 nodes, 3 degrees of freedom per node, 8 integration points in volume).
- For Shell Modelling: ELFORM = 16: Full integration with thin shell elements (4 nodes, 4 in-plane integration points, 5 through thickness), and a warping treatment algorithm (HGTYPE = 8) for out-of-plane loading.
- For Beam Modelling: ELFORM = 1: Hughes-Liu formulation with 4 Gauss integration points in the section

Drop of a Ball on an Aluminum Plate with a Dummy:

Junction: M5 bolts in M6 holes

Ball drops: 67g and 535g at 15cm and 45cm

Interfaces: 4 surfaces of 6 x 6 mm / Plane contact (80*80mm) / Steel washers / Permaglass washers

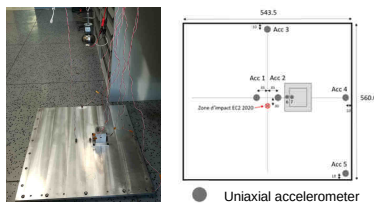


Figure 12: Illustrations of the Ball Drop Test on a Plate with a Dummy fixed on it and its instrumentation

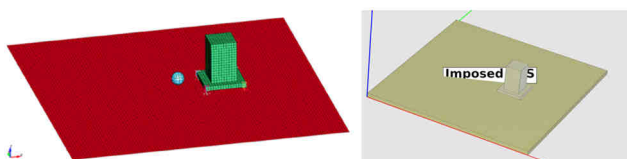


Figure 13: illustration of the Explicit FEM on LS-DYNA and TSEA on wave6 Modelling of the Studied Case

Description of the Explicit FEM Modelling:

Ball: ○ 535g ○ initial speed of 2.97m/s;

○ 024 Piecewise Linear Plasticity: Steel1

○ deformable to rigid after shock (3ms)

○ Solid Modelling: 4 elements on the radius

Plate: ○ 024 Piecewise Linear Plasticity: Alu1

○ 543.5 * 560 * 13 mm ○ element size: 5mm

Stacking Column Dummy:

○ 024 Piecewise Linear Plasticity: Alu2

○ 80 x 80 x 8 mm - 50 x 50 x 80 mm

○ element size: 5mm

Screws: ○ 024 Piecewise Linear Plasticity: Steel5;

○ Beam Modelling: 6 elements per screw from plate to dummy's top corners

Accelerometer modelling:

uniaxial for out-of-plane acceleration: None

Calculation Time Step:

At contact: 1.91e-5 ms - During oscillations: 2.39e-5 ms

Database Options DT:

○ GLSTAT; MATSUM; RCFORC; SLEOUT = 0.1ms

○ NODOUT = 0.0008 ms

○ SBTOUT = 0.01ms

○ BINARY D3PLOT=2ms

Control shell

Control bulk viscosity: Q1=1.5 Q2=0.7 TYPE=-2

Interface: 4 excess thicknesses 6*6mm

Contact Modelling:

Plate / ball: Contact automatic surface to surface

Plate / dummy: None

○ Pre-tightening Modelling: None

○ Joint Modelling:

○ Two nodal rigid bodies connected by a beam

○ Nodal rigid body - plate: 10x10 mm cross

○ Nodal rigid body - dummy: 6x6x13 mm

○ Resolution time: 18 minutes (16 CPUs)

○ Elements: 10812 shells / 2560 solids / 24 beam

Description of the Hybrid FEM-TSEA modelling:

Plate: ○ Linear Plasticity: Alu1 ○ 543.5*560*13 mm

○ Solid Modelling: TET4 / 5mm ○ Damping: 1%

○ Subsystem type: FE reverberant

Dummy: ○ Linear Plasticity: Alu2 ○ 80* 80*8 mm

+ 50*50* 80 mm ○ Solid Modelling: TET10 / 3mm

○ Damping: 1% ○ Subsystem type: FE reverberant

Screws: ○ Linear Plasticity: Steel5 ○ Damping: 1%

○ Beam Modelling ○ Subsystem type: FE

Joint modelling:

○ Two nodal rigid bodies connected by a beam

○ Nodal rigid body - plate: 10x10 mm cross

○ Nodal rigid body - dummy: 15x15x13 mm

○ Beam: 6 elements from plate to dummy's top corners

Shock Source Injection: Average of out-of-plane experimental SRS, injected over the entire plate

SRS calculation on plate nodes

Calculation of energies per subsystem

Solver Type: Time-Frequency

Modal Basis Calculation: AMS

One set of modes for all subsystems

Time Values: from 0 to 1 ms in 10 steps, then 2, 5, 8, 10, 50, 100 ms

Frequency Discretization: frequency label center ANSI 1/6 octaves up to 15 kHz

Resolution time: ~ 2 minutes

PSG on plate modelling

Description of the Explicit FEM modelling:

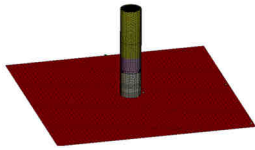


Figure 14: Illustration by LS-DYNA of the PSG Characterization on an Aluminum Plate

Plate: 024 Piecewise Linear Plasticity: Alu1

543.5 * 560 * 13 mm -- Shell Modelling: 5mm elements

PSG: Solid Modelling: 2mm elements

- Main Body PSG: Steel2
- PSG Chamber: Steel2 ○ PSG Anvil: Steel3
- Impactor: Steel4 ○ PSG Damper for T35: Ptfe
- M5 Bolt for Accelerometers: Steel2
- Triaxial Accelerometer Cubes: Alu3
- Screw M10 Bolt for Anvil/Plate: Steel2

Pressure curve injection

Accelerometer modelling:

○ Uniaxial for out-of-plane acceleration: 5mm x 5mm rigid shell element on the upper surface of the plate, each linked rigidly to five nodes in a cross pattern using *Constrained extra node set*.

○ Triaxial: Aluminum cubes connected to the plate via a flexible beam. Nodes for X, Y, and Z acceleration recovery aligned with experimental measurement zones

Elements:

12115 Shells / 65552 Solids / 20 Beams / 2 Mass Part / 9 Seatbelt Accelerometers

Resolution time: ~ 5 hours (8 CPUs)

Description of the Explicit FEM-TSEA Modelling:

Plate: Linear Plasticity: ○ Alu1 ○ 543.5 * 560 * 13 mm

Solid Modelling: TET4 / 5mm - Damping: 1%

Subsystem type: FE reverberant

Shock Source Injection: Average of out-of-plane experimental SRS, injected over the entire plate

Calculation of energies per subsystem

Modal Basis Calculation: AMS

Modal Resolution: one set of modes for all subsystems

Time Values: from 0 to 1 ms in 10 steps, then 2, 5, 8, 10, 50, 100 ms

Frequency Discretization: frequency label center ANSI 1/6 octaves up to 15kHz

Resolution time: ~ 1 minute

MicroCarb Satellite modelling

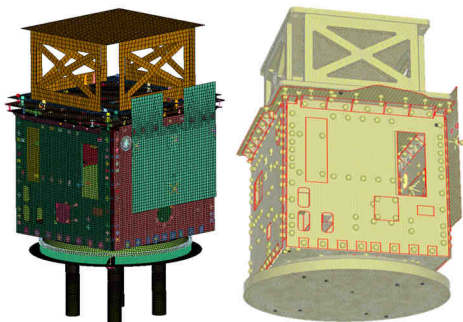


Figure 15: Illustration by LS-DYNA (left) for Explicit FEM and by wave6 for FEM/TSEA (right) of the MicroCarb structural model

Description of the Explicit FEM Modelling:

PSG: Same as “PSG on plate” case

Accelerometer modelling: Same as “PSG on plate” case

Elements: 116 Beam / 7 Mass / 10 Mass Parts / 41

Seatbelt Accelerometers / 62575 Shells 13671 Shell beta / 318596 Solid

Global damping: VALDMP=0.1

Database Options:

- NODOUT: DT = 0.0008 ms ○ RCFORC : DT=0.1ms
- SBTOUT: DT=0.01ms ○ SLEOUT: DT=0.1ms
- BINARY D3PLOT: DT=0.02ms

Contacts:

- PSG parts: Automatic single surface
 - Impactor – damper: Automatic surface to surface
- Resolution time: ~ 22 hours (16 CPUs)

Description of the Hybrid FEM/TSEA Modelling:

Injection Plate:

- Injection Plate Mesh:
 - Element size: 0.01m
 - Min. relative element size: 0.8
 - Min. relative feature size: 0.2
 - Preserve free edge: No
 - Preserve feature edge: Yes
- Preserve junction edges: No
- Material: Predefined Aluminum
- Damping: 1%
- Subsystem Type: FE reverberant

Remaining Subsystems:

Meshes, materials, and damping are defined by the FEM

Subsystem Type:

- FE reverberant if SRS measurement
- FE otherwise

Connection between the Injection Plate and Payload

Adapter: Linear junction between the injection plate and the payload adapter on bolted surfaces.

Other Connections: Beam and nodal rigid body connections typically used in FEM vibration models.

Shock Source Injections:

○ Average of out-of-plane experimental SRS, injected over the entire injection plate.

○ Normalized and unnormalized frequency energy distribution derived from characterization of the shock source on an aluminium plate (previous paragraph configuration), using modelling parameters identical to those in the previous paragraph.

Calculation of SRS on nodes corresponding to accelerometers along each axis of the global reference frame

Solver Type: Time-Frequency

One set of modes for all subsystems except masses (AMS calculation with 15 kHz limit)

One set of modes for masses (Calculation limited to 5 modes below 5 Hz)

Temporal discretization: from 0 to 0.001s in 10 steps, then 0.002, 0.005, 0.008, 0.01, 0.05, 0.1

Frequency discretization: frequency label center ANSI 1/6 octaves

Frequency study limit: 15 kHz

Number of Elements: Same as FEM

Resolution time: ~ 9 hours (16 CPUs)